



Discovering regularity and mechanisms of word sense acquisition in childhood

Jiangtian Li^{a,1} , Blair C. Armstrong^{a,b} , and Yang Xu^c

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How does language use inform the emergence of word meanings in early life? Prior work in developmental psychology and the cognitive sciences typically focuses on studying word acquisition in children without specifying how different senses within a word emerge through time. To shed light on word sense acquisition, we propose a framework grounded in state-of-the-art computational methodologies of contextual word embedding to characterize how different senses of a word unfold as children acquire their lexicon. Our framework identifies word senses automatically by forming semantic clusters through natural language use, and analyzes 1,270 words from approximately 4 million utterances produced by children (19 to 144 mo) and their caretakers. The psychological validity of these senses was assessed based on a combination of dictionaries, human judgment of semantic similarity, and evaluation from a large language model. We then tested three hypotheses motivated by existing work on word sense emergence in language evolution. First, concrete senses of a word tend to emerge earlier than more abstract senses in child language. Second, word senses grow incrementally in semantic space across development. Third, algorithms of semantic chaining—how words spawn new senses by extending from existing senses—recapitulate the order of word senses in development. We find support for all three hypotheses. Our work suggests that the development of word meaning in language acquisition resembles that in language evolution and offers a converging view on the cognitive principles and mechanisms across timescales in the ontogeny and phylogeny of word sense emergence.

word meaning | polysemy | language development | word sense acquisition | contextual word embedding

There are more meanings than word forms in the lexicon, a phenomenon commonly known as polysemy or lexical ambiguity (1–4). In language development, a child initially acquires the first meaning of a word, which often bears minimal relationship to the word form due to sound–meaning mappings that are largely arbitrary (5). As the child's lexicon expands over time, they learn to map multiple meanings onto a word form. Although there is extensive literature on how children learn the first meaning or a single meaning of a word (6–11), less is understood about how polysemous meanings of a word, or “word senses,” are acquired at scale through developmental time.

In this study, we investigate how different senses of a word emerge in childhood and what general mechanisms might underlie the acquisition of polysemous word meanings using a computational framework applied to naturalistic child speech data. Fig. 1 illustrates this process, which we call “word sense acquisition” (abbreviated as WSA). For example, the word “stick” has the senses of a thin piece (wooden stick), to thrust (stick out a finger), and to adhere (stick to the wall), which children generally learn in this order. To better situate our work, we first review relevant work in two areas: a) computational modeling of polysemy and its historical evolution, and b) child lexical acquisition.

Considerable progress has been made in the computational modeling of polysemy and the evolution of word meaning. Recent models of polysemy are often based on the use theory of meaning (12, 13) and distributional semantics (see ref. 14 for review); that is, the meaning of a word is reflected in its use or the words it co-occurs with in a given linguistic context. In particular, contextual word embedding models, such as transformer-based BERT (15), have been used effectively to capture polysemous word senses both in terms of the variety of senses and the relationships among them (16–19). These models can produce context-sensitive representations of word meanings in a reliable and scalable fashion (20–22), bypassing a big challenge in studying polysemy: classifying different uses in raw texts into distinct word senses. Despite this progress, many studies present a static view of the lexicon focusing on the adult population, without characterizing how

Significance

Many words in natural language have more than one meaning. For example, “stick” can refer to the senses of a thin piece, a thrusting motion, or the state of adhering. Children as language learners acquire these different senses of a word in language development, but it is not well understood how they acquire them and in what order. In this study, we use computational methods that draw on models from artificial intelligence to identify regular patterns in word sense acquisition through developmental time. We find that children's acquisition of word senses shares similar cognitive principles and mechanisms with those found previously in the historical emergence of word senses.

Author affiliations: ^aDepartment of Psychology, University of Toronto Scarborough 1265 Military Trail Toronto, ON M1C 1A4, Canada; ^bBasque Center on Cognition, Brain, and Language, Paseo Mikeletegi 69, 2º, Donostia - San Sebastián 20009, Gipuzkoa, Spain; and ^cDepartment of Computer Science, Cognitive Science Program, 6 King's College Road, Toronto, ON M5S 3H5, Canada

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¹To whom correspondence may be addressed. Email: jiangtian.li@utoronto.ca.

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word senses emerge during childhood. One exception is related work on the historical evolution of word meaning, wherein researchers have applied computational models to characterize how novel word senses emerge through historical periods (23–25). However, such an approach has not been taken to model the emergence of word senses at scale in child language development.

Separate work on child word acquisition has examined how words become associated with meanings in development. For example, children learn word meanings based on how words are used by others when communicating (7). Existing work has typically made the simplifying assumption that words have a single meaning (e.g., refs. 8 and 9), whereas research on lexical ambiguity has established that more than 80% of words are ambiguous and take on different senses in different contexts (1). For example, as illustrated in Fig. 1, the different senses of “stick” may be expressed by and to children at different ages. Prior work has examined the acquisition of different word senses during child development (26–30), making important empirical observations on how children acquire multiple word senses. However, these studies often focus on a small set of words restricted to particular parts of speech (29, 30) or exhibiting specific patterns of polysemy (e.g., the animal/meat distinction as in chicken and lobster) (26–28), or they rely on resource-intensive manual tagging of word senses (31). These scalability issues, both in terms of studying larger sets of words and tagging their senses, could potentially be addressed with recently developed computational tools. Although progress has been made in applying recent computational developments to study children’s word acquisition [e.g., integrating a wide variety of cross-linguistic datasets (10), using word embeddings (32), or using semantic network models (11)], this line of work stays at the level of words, hence it does not capture the development of internal sense structures *within* each word. Therefore, our current study integrates these computational methods with large-scale child speech corpora but applies them at the level of word senses, which currently lacks systematic and large-scale investigation.

We present a computational framework to characterize children’s WSA by considering a variety of parts of speech (e.g., nouns, verbs) and using longitudinal transcribed data from child speech. To our knowledge, this is the first comprehensive analysis on the early emergence of word senses in the child lexicon, as opposed to assuming one word meaning per word and studying learning at the level of words (33). Our work connects a number of disciplines in cognitive science, starting from the philosophy of language (particularly use theory of meaning), and expanding to include linguistic work on historical emergence of word senses, psychological work on word acquisition and polysemy, and formal computational models from artificial intelligence.

Utilizing this framework, we examine three hypotheses about the regularity and mechanisms of WSA. These hypotheses are motivated by existing work on historical word sense emergence, and shed light on whether similar principles and mechanisms are at play in a development context. The hypotheses, which are illustrated in Fig. 2 can be summarized as follows:

- Hypothesis 1: Concrete senses of a word tend to emerge earlier than more abstract senses in child language development (cf. 24, 29, 30, 34, 35).
- Hypothesis 2: Word senses grow incrementally in semantic space over developmental time (cf. 36–39).
- Hypothesis 3: Mechanisms of semantic chaining recapitulate the order of emergence of word senses in child development as words spawn new senses from existing ones (cf. 23, 25, 40–43).

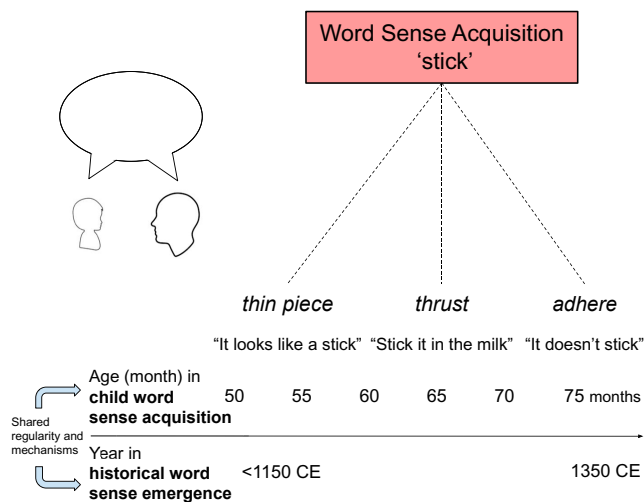


Fig. 1. An illustration of the problem of word sense acquisition in child language development, and its potential links with word sense emergence in historical language evolution.

Computational Framework for Word Sense Clustering

Our framework aimed to advance our understanding of WSA by overcoming several key limitations of past work, which has relied upon human annotation of senses. First, human annotation by experts is resource-intensive, which gives only a small and potentially nonrepresentative glimpse into the lexicon (29, 30). Second, there are no standard sources of word senses that cover child development. Children could acquire a partial set or even a different set of senses from adults, which makes standard sources of word senses, such as dictionaries (44) and databases (45), less useful as sources of constraint for this problem, although child-focused dictionaries are an underexplored resource on this front (e.g., ref. 46). Because of both the aforementioned reasons, crowd-sourcing word sense identification by (adult) humans (31) is a useful but limited way forward. Although more data can be analyzed, it is still resource-intensive, and the more diverse prior linguistic knowledge and expertise of raters adds additional noise to such endeavors (47).

We overcame these issues encountered in past work by extracting sense-like clusters from child-relevant speech from the CHILDES corpus (48), which includes both child-directed speech and child speech. First, we generated contextual embeddings for a target word in CHILDES using the established pre-trained word embedding model BERT (15). Next, we conducted unsupervised semantic clustering (49) of word meanings into sense-like clusters in child-relevant speech from the CHILDES corpus based on these embeddings. Finally, with such clusters in hand, the properties of these senses, such as sense similarity, sense concreteness, and sense production age, can be calculated to enable the examination of various hypotheses about WSA. The details of the implementation are included in *Materials and Methods*.

The first main advantage of this approach is that it allowed us to bootstrap the semantic information from child-directed speech to inform the sense categories within child speech (which is examined separately from the child-directed speech in our analyses below). This approach compensates for the data sparsity associated with child-directed and especially child speech that might otherwise limit such a computational approach—although CHILDES is a large corpus of child-relevant speech, at only

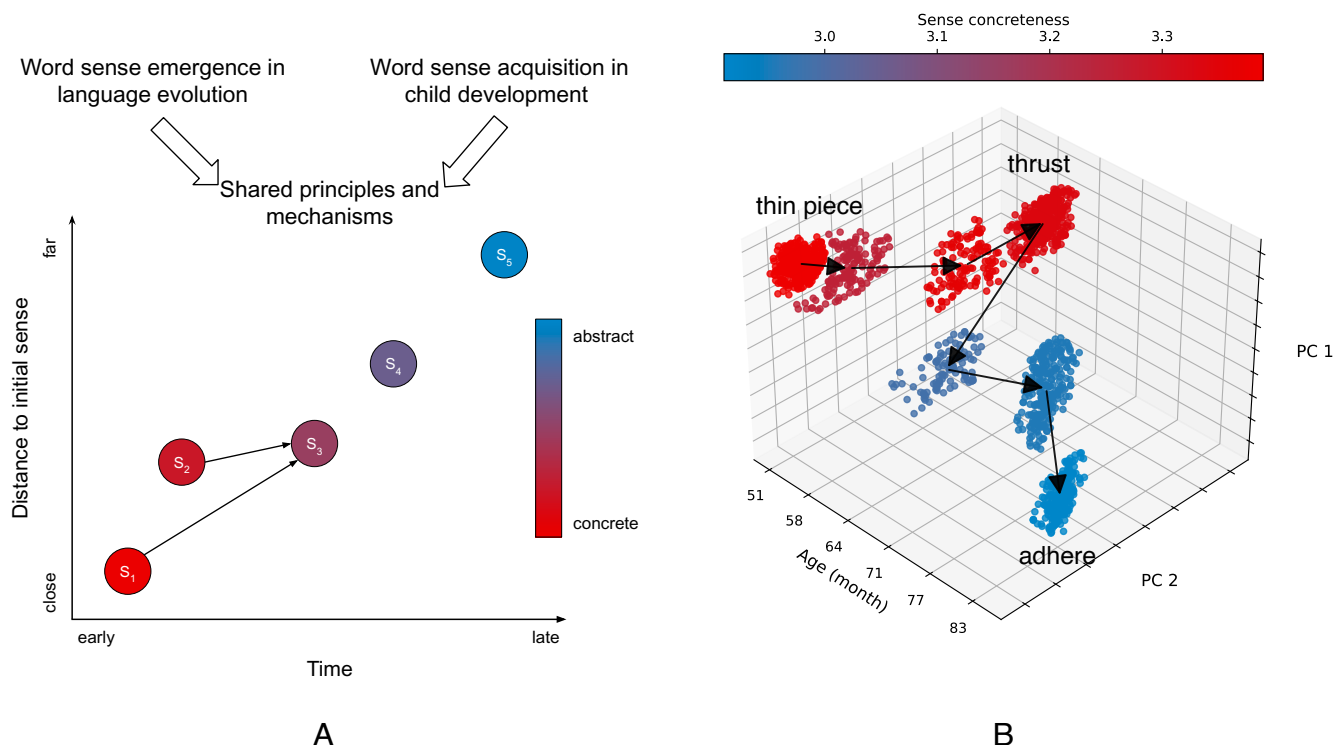


Fig. 2. Summary of the framework for investigating word sense acquisition. (A) Illustration of the three hypotheses for word sense emergence shared across language evolution over historical time and word sense acquisition in child development: (1) Sense extension is directional, from concrete to abstract senses; (2) New senses emerge in the semantic space adjacent to previously learned senses; (3) A new sense (e.g., S_3 in the figure) is acquired through the mechanism of being chained to existing senses that are conceptually related (e.g., S_1 and S_2 in the figure). (B) Three-dimensional visualization of the joint sense clustering of “stick” in child and child-directed speech through developmental time. Two of the dimensions depict the first two principal components of the semantic space from the BERT model, discussed later, and the third dimension depicts the mean sense age in child speech. The color of the cluster denotes sense concreteness.

14 million word tokens, it is still a relatively small corpus. Thus, successful evaluation of our hypotheses also answers the overarching question of whether such a small source of natural language data is sufficient to probe finer-grained details related to word senses. The second main advantage was that the use of an unsupervised clustering approach, specifically a nonparametric Dirichlet Process Gaussian Mixture Model (49), allowed us to infer an appropriate number of sense clusters based on the distribution of the data without relying upon a separate standard source for coding word senses, or even how many senses should be extracted (cf. 18). It also allowed us to leverage a clustering approach that has proven robust to detecting structured clusters in studies of other aspects of human cognition, such as incremental categorization (50) and argument structure acquisition (43, 51).

Results

Psychological Reality of the Word Sense Clusters. We derived semantic clusters for 1,270 words in the CHILDES corpus, which essentially exhausted the clusterable lexicon in CHILDES (North American English) for which we had a minimum of 200 utterances in the corpus. The mean number of utterances for each word was 3,603, and the mean number of sense clusters per word was 6 (SD = 3.4), ranging from 2 to 28. The mean age of each sense in child speech was 53 mo (SD = 13), ranging from 19 to 144 mo. The mean sense age in child-directed speech was 41 mo (SD = 9), ranging from 6 to 94 mo. This latter finding is consistent with the earlier emergence of comprehension as compared to the production of a target word (52).

A visualization of the clustering for one target word, “stick,” is presented in Fig. 2B, with individual points depicting the context vectors for utterances in the first two principal components of the PCA and mean sense age in child speech. In this visualization, senses are ordered in the third dimension of age. We observe that the sense of “thin piece” emerged in child language prior to the sense of “to thrust” and “to adhere.” Upon inspection of the individual clusters for “stick,” it is clear that the sense clusters are fine-grained, so one dictionary sense could encompass several sense clusters. Table 1 provides example utterances.

Among the 1,270 words, 1,021 are polysemes (i.e., words with related senses), 157 are homonyms (i.e., words with unrelated senses), and 92 are monosemes (i.e., words with only one sense) according to the Wordsmyth Dictionary, which provides broad coverage of the (adult) lexicon. To examine children’s acquisition of different kinds of lexical ambiguity, we compared the relatedness of sense clusters induced for homonyms vs. polysemes. We calculated the maximal semantic distance (measured using cosine distance) across all pairs of sense clusters for each word. These maximal differences did not significantly differ between homonyms ($M = 0.255$, $SE = 0.008$) and polysemes ($M = 0.248$, $SE = 0.003$; $t(191) = 0.83$, $P = 0.41$), a point we return to in the discussion.

After deriving the word sense clusters, we conducted three sets of validation to examine the psychological reality of the sense clusters identified by our computational framework. The first validated the psychological reality of the sense clusters vis-à-vis sense counts from dictionary definitions. The second validated the psychological reality of the sense clusters in a psycholinguistic experiment. The third evaluated each sense cluster with a large language model.

Table 1. The sense clusters identified in the computational framework for the word “stick,” as well as their mean age in child speech, randomly selected example utterances from a child (CHI) or their parent (MOT), Wordsmyth Kids definitions aligned by GPT, GPT-generated definitions, and GPT-rated homogeneity

Senses	Age (months)	Example utterances	Wordsmyth definition	GPT definition	Homogeneity (%)
Stick ₁	51	CHI: Baby. There's a stick in there. There's a stick in there. MOT: This one has a stick. This one has a ball on his head. This one has a ball on his head.	A stick is a long, thin piece of wood.	Noun: a long thin piece of wood or similar material (e.g., a stick used for walking, playing, hitting, or support).	95
Stick ₂	53	CHI: Lipstick under there. There. Stick. There. Stick. Kipstick. There. Fish. Fish. Fish. MOT: Well ‡ we can put it in. Yeah. Mom can put the stick in. We just can't put any air in. Here ‡ let me have it.	A stick is a long, thin piece of wood.	Primarily noun: a (usually wooden) short thin piece used as an object (popsicle stick, measuring stick, pretzel stick) and related toy parts; some verb uses but mostly noun.	75
Stick ₃	59	CHI: An put my name up there and will stick up there. MOT: Abe Daddy can throw you up on the ceiling. So you can stick on the big piece of paper. And I could get a marker and mark around you.	none of the Wordsmyth definitions fits the full variety of idiomatic uses in this cluster.	Mixed/idiomatic uses: various senses including “stay nearby” (stick around), “adhere,” “insert/push in,” “protrude/stick out,” and idiomatic “stick with” (continue)—cluster is heterogeneous.	30
Stick ₄	63	CHI: Okay. Hey you hafta be daddy o. No, he he the. Make believe like. Timothy. Hey Timothy. Stick them in the, in my back. Something's in my pocket. Hey you wanna play with this? Barry wanna play with it or not? Six million dollar man. I. Okay make believe like he stole your car. MOT: So put the r here. Stick it in there. Read it back to Mommy.	When you stick something in place, you push the pointed end of it into something.	Verb (mixed): to put or insert something into or onto something (including pushing a pointed end into something, placing stickers, or sticking items together); cluster mixes insertion and attachment senses.	60
Stick ₅	66	CHI: Don't these stick? MOT: Okay. Okay. Take it out this way. Slide this out. Careful. I think these are magnets. Amelia, look, they're sticky. They stick. And then you can get all the animals out. Wow what is this?	When something sticks, it means that it becomes attached to something. When a feather sticks to your shirt, it becomes attached to your shirt.	Verb: to become attached or stuck to something (includes magnets, adhesives, or prickly attachment like cactus); also used for describing things that stay stuck together.	85
Stick ₆	73	CHI: Yeah because it's an opposite side. Watch. I don't think these will stick to this side. Oh! They do. Hey where's that thing? MOT: That's a dime. There's a dime and a nickel. Will they stick to the magnet?	When something sticks, it means that it becomes attached to something. When a feather sticks to your shirt, it becomes attached to your shirt.	Verb: to become attached or adhere to a surface (including magnetic attraction); also used for things being sticky or clinging together.	90
Stick ₇	83	CHI: Nanny. Mine doesn't stick. MOT: I'm pretty sure that those go on there. Put the zebra on there. Look. Look! Oh maybe it doesn't stick. Oh it does stick. See? Zebra goes here. What else you wanna play with? What's that? What do you have?	When something sticks, it means that it becomes attached to something. When a feather sticks to your shirt, it becomes attached to your shirt.	Verb: to adhere or be held in place (often used in contexts of magnets or glue—frequently negative: “doesn't stick”).	90

In the first validation, we compared the number of sense clusters for each word with its corresponding number of senses in three dictionaries: Wordsmyth-Kids (46), Wordsmyth (44), and WordNet (45). The correlations between the number of senses for the 1,098 words that are shared across these 3 dictionaries were all significant (Wordsmyth-Kids: $r(1096) = 0.25$, $P < 0.001$, Wordsmyth: $r(1096) = 0.22$, $P < 0.001$, WordNet: $r(1096) = 0.13$, $P < 0.001$) and highest in the Wordsmyth-Kids dictionary, presumably because it included the most relevant senses used among children. As context for the correlations between the number of sense clusters and the dictionary sense counts, it is worth noting that there is considerable variability across dictionaries themselves. For example, the correlation between Wordsmyth and WordNet is 0.72 ($df = 1,096$, $P < 0.001$). Thus, finding a correlation close to 1.0 between our measure and dictionaries is not expected. The overall correlations and the correlations computed separately for words based on their dominant POS across all utterances, nouns, verbs, and adjectives (the three most frequent POS) are presented in Fig. 3A.

In the second validation, we conducted a preregistered psycholinguistic experiment to examine the psychological reality

of the sense clusters. The full details regarding this experiment are presented in SI Appendix. Briefly, we collected semantic similarity ratings for pairs of utterances containing the same target word, with each utterance drawn from a sense cluster. They could come from the same sense cluster (Cluster A and A), different sense clusters that were close to one another (Cluster A and C), or different sense clusters that were distant from one another (Cluster A and B). Examples of sense clusters and utterances are provided in Table 2. The overall similarity levels were relatively high in all three conditions, which is attributable to two considerations. First, the majority of words are polysemes that have related senses, so their senses are expected to be quite similar. Second, our computational approach identifies fine-grained senses clusters which are often quite close to one another and reflect similar meanings. Nevertheless, participant similarity ratings exhibited the expected graded effect across conditions, with similarity ratings decreasing from the same cluster (AA) condition ($M = 5.48$, $SE = 0.02$) to the close clusters (AC) condition ($M = 5.02$, $SE = 0.03$) to the far clusters (AB) condition ($M = 4.64$, $SE = 0.03$), with ratings on a 1-7 Likert scale (1 = very different meanings, 7 = very similar meanings)

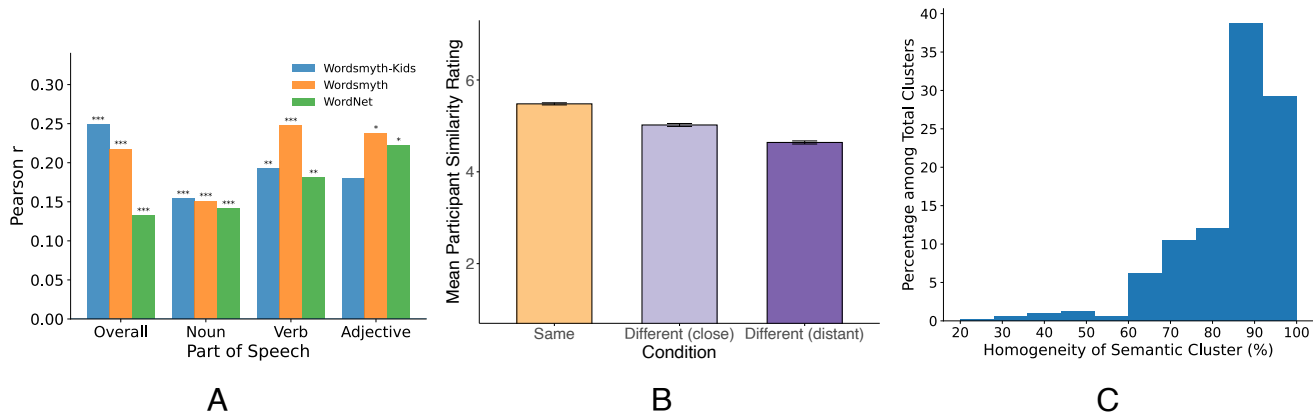


Fig. 3. Validation of the sense clusters induced by our computational framework. (A) Correlation between the number of sense clusters induced by our computational framework and the number of senses (NOS) in three dictionaries: Wordsmyth-Kids, Wordsmyth, and WordNet. The latter two sources are reflective of the broader adult lexicon. Each group of bars represents the correlations for all words combined ($n = 1,098$), as well as separate correlations for words based on their dominant POS across all utterances: nouns ($n = 582$), verbs ($n = 219$), and adjectives ($n = 106$). $*P < 0.05$; $**P < 0.01$; $***P < 0.001$ —statistical significance levels. (B) Bar plot of participants' similarity ratings for two utterances from each of the three conditions: same sense cluster, different sense clusters that are close to one another, and different sense clusters that are distant from one another. Error bars depict SEs. (C) Sense homogeneity within each sense cluster for 7,645 sense clusters across 1,270 words.

(Fig. 3B). We conducted linear mixed-effects analyses and found that all three conditions were significantly different from each other ($P < 0.001$ after correcting for multiple comparisons) (additional details are reported in *SI Appendix, section 1* including example sentences in *SI Appendix, Table S1*).

In the third validation, we used a state-of-the-art large language model, GPT-5 mini, to evaluate the quality of word sense clusters identified in our framework. For each sense cluster identified by our computational framework across all of the 7,645 sense clusters from 1,270 words we analyzed, we randomly sampled 400 utterances (or utilized all available utterances for clusters with fewer than 400 utterances). We then fed these utterances into GPT-5 and asked it to 1) provide a definition for each sense, 2) indicate the homogeneity of the sense cluster, that is, the proportion of utterances that represented the same sense, 3) to indicate if a sense definition in Wordsmyth Kids matched the sense usage in each cluster, and 4) if such a definition existed, to identify that definition. GPT-5 was able to provide definitions for each sense cluster, and the homogeneity of the sense clusters was generally high ($M = 84\%$, $SD = 12\%$), with more than 75% of the senses achieving a homogeneity of 80% (Fig. 3C). For the majority of the senses (92%), GPT-5 was able to find a matching definition from Wordsmyth-Kids (see Table 1 for examples of

definitions). Manual inspection of the 8% of cases for which a definition could not be found in Wordsmyth-Kids suggested that these correspond to child-specific senses (e.g., “muffin” in “the Muffin Man”) and phrasal constructions (e.g., “god” in “oh my god”) (see *SI Appendix* for additional details).

In summary, the sense clusters induced by our computational framework are psychologically realistic in that the sense clusters it identifies reflect established variability in the number of senses associated with a given word, the distinction between different word senses established via human subjective ratings, and the homogeneity of language use within a sense cluster. Furthermore, the majority of the senses can be matched to definitions in dictionaries and those that cannot have sensible interpretations in a child-specific context.

Hypothesis 1: Concrete-to-Abstract Emergence of Word Senses. We tested our first hypothesis by examining whether there is systematicity in the directionality of WSA, that is, whether there are properties of senses that impact the timing of their acquisition during development. Specifically, we examined whether concrete senses tend to be acquired earlier than more abstract senses.

It is known that children acquire concrete words earlier than abstract words (33, 53) when researchers have made the

Table 2. Example utterances used in participant sense similarity rating tasks for the word “cover.”

Sense cluster	Interpretation	Speaker	Example utterances
A	Hide	Child	“And we hafta COVER him up.”
		Child	“COVER up Captain Hook. Let’s throw him away.”
		Caregiver	“COVER it up. Make it a present.”
		Caregiver	“COVER the bird up.”
B	Object that lies over	Child	“I’m going to get my COVER!”
		Child	“Pull the COVER off.”
		Caregiver	“You wanna put the COVER on?”
		Caregiver	“It needs the COVER on Abe.”
C	Put on top	Child	“Did you COVER your watch?”
		Child	“Because I COVER her.”
		Caregiver	“Don’t hold it and it’ll go round. See watch it spin. Let me see if it spins. Hey look at it go round. Do you think you’re going to wake your baby up? Oh we better COVER her and put her to sleep.”
		Caregiver	“The box. Get the other box and I’ll COVER this one. Go ahead.”

simplifying assumption that words are only associated with a single sense and thus presumably are making this claim based on the first sense to be learned for each word. Here we hypothesize how this effect manifests as children acquire individual senses for a given word. Specifically, we propose that children first typically acquire more concrete senses of a given word (i.e., the base sense), which are then extended to more abstract senses (i.e., the extended senses) (depicted in Fig. 2A). This hypothesis has garnered support from studies on how polysemy (such as metaphorical word meaning) emerges over historical time (24, 34), and has been cross-validated in hundreds of languages (35). However, the applicability of this principle to the case of developmental sense acquisition has been questioned by refs. 29 and 30, who provided evidence that the senses of prepositions acquired early in development are not necessarily reflective of the historical pattern of word sense extension. We were motivated to reexamine this inconsistency in the concrete-to-abstract emergence in WSA among other parts of speech and a large set of words.

As a result, if sense concreteness does shape the directionality of WSA, we expect a negative relation between sense concreteness and the mean age at which a sense appears in child speech. Overall, sense concreteness had a mean of 3.3, a SD of 0.26, and ranged from 1.7 to 5.0 in our dataset. We also included two control variables in our analysis: log sense frequency and noun proportion, as previous literature (29, 54) suggests that they may also influence the age of acquisition (detailed definitions of these variables are provided in Fig. 4A).

We used mixed-effects regression models to predict the average age of each sense in child speech with three fixed effects: 1) sense concreteness, 2) log sense frequency in child-directed speech, and 3) noun proportion. Besides the regression model on the sense from all words in the lexicon, we also fitted 3 separate models for senses among nouns, verbs, and adjectives, which are the most frequent parts of speech categories (models for other parts of speech are reported in SI Appendix).

With respect to our hypothesis, we found a significant negative effect of sense concreteness (standardized $\beta = -0.09$, $se = 0.02$, $t(859.37) = -6.23$, $P < 0.001$) in predicting the average sense age in child speech when including all parts of speech. The results are presented in Fig. 4B and C. Focusing on the three most frequent part of speech categories, all of them are significantly negative (nouns: standardized $\beta = -0.06$, $se = 0.02$, $t(481.70) = -2.45$, $P = 0.015$; verbs: standardized $\beta = -0.12$, $se = 0.04$, $t(162.59) = -3.26$, $P = 0.001$; adjectives: standardized $\beta = -0.14$, $se = 0.06$, $t(69.81) = -2.44$, $P = 0.017$). Overall, these results indicate that children tend to produce more concrete senses at younger ages.

We also found a significant negative effect of noun proportion for words overall (standardized $\beta = -0.16$, $se = 0.02$, $t(867.16) = -8.33$, $P < 0.001$) and verbs (standardized $\beta = -0.07$, $se = 0.03$, $t(68.35) = -2.26$, $P = 0.027$), which replicated previous findings that noun senses tend to be acquired earlier than other senses (54). We did not find significant effects of sense frequency but we found a positive effect for verbs (standardized $\beta = 0.08$, $se = 0.03$, $t(1329.21) = 2.88$, $P = 0.004$). This is due to the fact that higher-frequency senses could span a larger range of ages and dilute their density within a single cross-section. When we controlled for the sense age span, the sense frequency effects all turned significantly negative, as detailed in SI Appendix, section 4.

We also conducted an analogous analysis predicting the average sense age of the children to whom speech is directed (i.e., child-directed speech), which examines whether caretakers also produce word senses to children following a concrete-to-

abstract order. We did not find evidence for such directionality in this analysis (standardized $\beta = 0.02$, $se = 0.02$, $t(819.04) = 1.11$, $P = 0.266$). This indicates that child sense emergence is governed by constraints intrinsic to children and children are not simply recapitulating the senses presented to them by caregivers at different ages. The details of this analysis are presented in SI Appendix, section 5.

Hypothesis 2: Incremental Growth of Word Senses. Next, we examined whether the sense clusters produced by children expand incrementally over developmental time.

In historical semantics, existing research has suggested that new word senses do not emerge randomly in historical time during language evolution, but they tend to develop in incremental ways that occur in adjacent portions of semantic space (36, 37). Recent computational work on semantic change also offers evidence that word meanings tend to shift smoothly over time (38, 39). As illustrated in Fig. 2A, the way in which new senses grow over time may lead to incremental expansions of the semantic space, such that the second emerging sense may appear semantically close to the first sense, and the third emergent sense is semantically close to the second sense, and so on. However, there is currently no comprehensive analysis examining the incremental growth or expansion of word senses in the child lexicon.

Insofar as there is incremental growth of word senses, we hypothesize that semantically similar senses emerge close in time during WSA, and conversely, that less similar senses will emerge at more disparate times during development. In other words, we expect a positive relationship between pair-wise sense distance and pair-wise age difference.

To this end, we conducted regression analyses by fitting mixed-effects models with all the pair-wise combinations of sense clusters for every word. This model predicted the age difference between each pair of senses of a given word in child speech from two fixed effect predictors: 1) sense distance and 2) log sense frequency difference. Detailed definitions for these variables are provided in Fig. 4A. This second variable was included to control for base effects of frequency. Similarly to how we test Hypothesis 1, we fit one regression model for all words in the lexicon, as well as separate regression models for each major part of speech (models for other parts of speech are reported in SI Appendix).

The estimated coefficients of the fitted models are provided in Fig. 4D and the relationship between sense distance and age difference is visualized in Fig. 4E. Critically, as predicted, there was a significant positive effect of sense distance in predicting the age difference between sense clusters when considering all words together (overall) or analyzing words from the three most prevalent parts of speech separately (all words: standardized $\beta = 0.28$, $se = 0.01$, $t(594.09) = 25.78$, $P < 0.001$; nouns: standardized $\beta = 0.26$, $se = 0.02$, $t(347.55) = 15.44$, $P < 0.001$; verbs: standardized $\beta = 0.28$, $se = 0.02$, $t(122.54) = 12.53$, $P < 0.001$; adjectives: standardized $\beta = 0.21$, $se = 0.04$, $t(53.20) = 5.24$, $P < 0.001$); even when the (significant) effect of log sense frequency difference is taken into account. These findings indicate that incremental sense growth is a guiding principle of child sense acquisition both for the whole lexicon and for the major part of speech categories.

Hypothesis 3: Semantic Chaining of Word Senses. Related to the incremental growth hypothesis, we tested the possibility that mechanisms of semantic chaining underlie child WSA.

Work on historical word sense emergence has proposed cognitive mechanisms whereby new senses emerge in connection

Variable type	Variable name	Definition
Response	Sense age difference	The log-transformed absolute value of the difference between the mean age in child speech for a given pair of sense clusters.
	Sense distance	The semantic distance between the two sense clusters in a given pair.
Predictors	Log sense frequency difference	The absolute value of the difference between the log sense frequencies of the two sense clusters in a given pair.
	Sense age	The mean age of utterances in child speech for a given sense cluster.
	Sense concreteness	The average concreteness associated with the utterances for a given sense cluster.
	Log sense frequency	The log frequency of the sense in the child's environment (child-directed speech).
Response	Noun proportion	The proportion of utterances in which the target word is used as a noun, among all utterances for a given sense cluster.

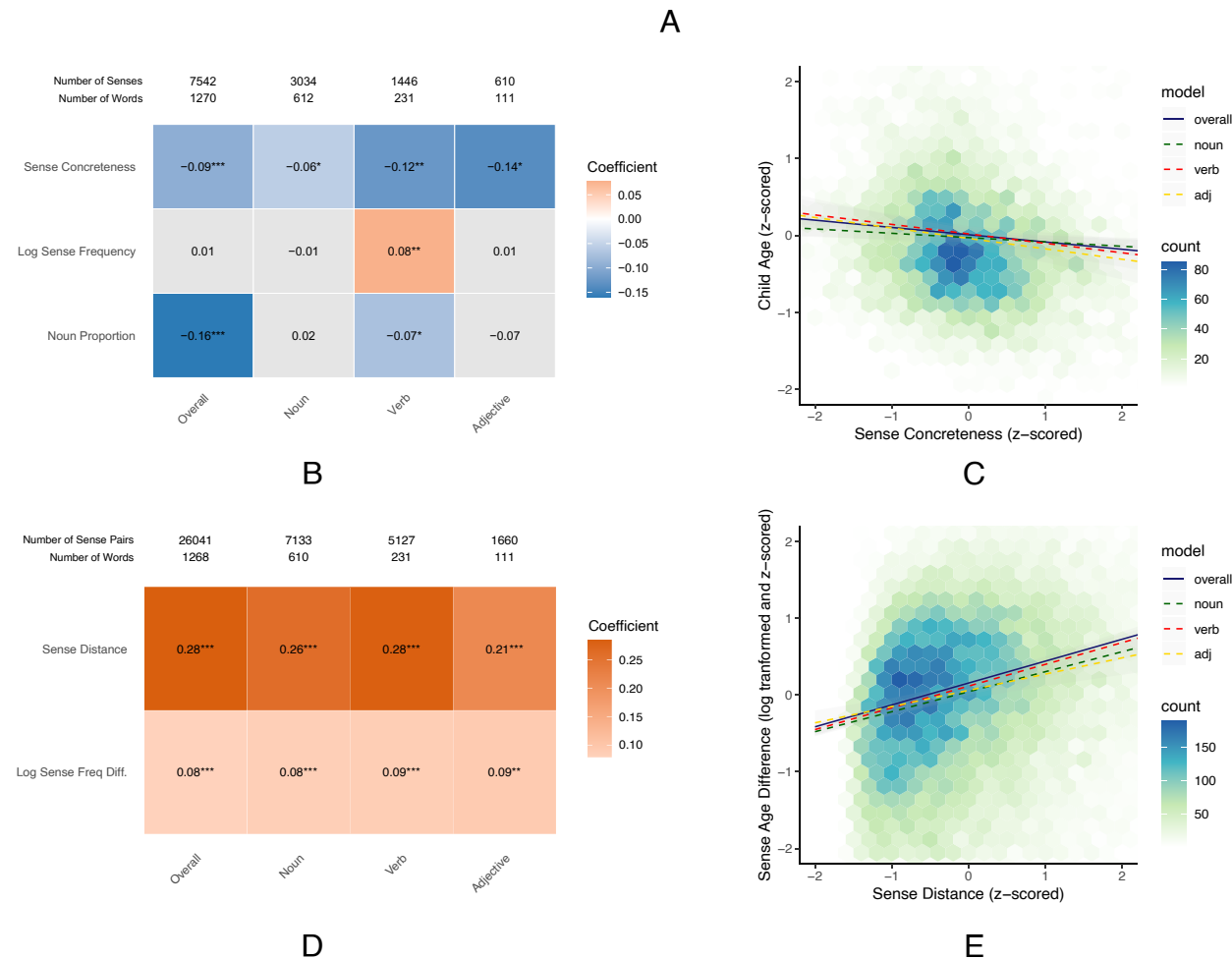


Fig. 4. The results of examining the hypothesis of and systematic directionality (Hypothesis 1) and incremental sense extension (Hypothesis 2). (A) The response and predictors used in the regressions analyses for the two hypotheses. (B) Systematic directionality: estimated coefficients for predicting mean sense age in child speech for all words, and words based on their dominant POS across all utterances: nouns, verbs, and adjectives. (C) Systematic directionality: effect plot of sense concreteness (z-scored) in predicting the mean sense age in child speech (z-scored) for all words, nouns, verbs, and adjectives, with the ribbon representing the CI and the hexbin layer shows the density of the data points. (D) Sense incrementality: standardized coefficients in predicting the pair-wise age difference between sense clusters for words and words based on their dominant POS across all utterances: nouns, verbs, and adjectives. (E) Sense incrementality: effect plot of sense distance (z-scored) in predicting the pair-wise sense age difference (log transformed and then z-scored) in child speech for all words, nouns, verbs, and adjectives, with the ribbon representing the CI and the hexbin layer shows the density of the data points. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$ correspond to different levels of statistical significance. The backgrounds of nonsignificant coefficients are grayed out.

with existing senses. Motivated by cognitive-linguistic theories of semantic chaining (40, 41), prior studies explored several computational algorithms to reconstruct the historical emergence of word meaning. In particular, a class of chaining algorithms was developed (25) to reconstruct the order of sense emergence according to a historical dictionary. It was found that dictionary-based senses are best recapitulated by “Nearest neighbor chaining” where a new sense emerges by linking to an existing sense most proximal in semantic space (see also ref. 23). Follow-up

work showed that “Exemplar chaining” better accounts for historical semantic growth in natural corpora and that Nearest neighbor chaining is a special case of exemplar-based meaning extension (42, 43). Initial support that similar exemplar-based mechanisms are at play in the case of developmental WSA has been reported for spatial terms only (55). Here, we apply the existing framework of semantic chaining from ref. 25 to explore whether these mechanisms also account for the order of sense emergence in the child lexicon.

Following ref. 25, we focus on five different algorithms specifying how a new sense emerges by linking to particular existing senses of a word as follows: 1) Progenitor chaining: linkage to the earliest existing sense; 2) Local chaining: linkage to the latest emerged sense; 3) Nearest neighbor chaining: linkage to the most similar existing sense; 4) Exemplar chaining: linkage to all existing senses proportional to their semantic similarities, and 5) Prototype chaining: linkage to the prototype of the existing senses. We provide the detailed formalisms of these algorithms in *Materials and Methods*.

Insofar as semantic chaining hypothesis is true, we expect that all five semantic chaining algorithms (Exemplar, Prototype, Progenitor, Local, and Nearest neighbor) better predict the temporal sequence of word senses acquired by children in comparison to a random baseline model, which arbitrarily predicts the order in which a word's senses are acquired by children. We evaluated these chaining algorithms using the same procedure used by ref. 25, who calculated the log likelihood of the sequence of emerging senses for all five chaining models, over the 1,270 words and their associated senses that form our dataset. For each word, we compared the predictions of these five semantic chaining models against the baseline model by calculating the log likelihood ratio. The results are presented in Fig. 5, and reveals that the log likelihood ratios of all five models against the random model were all significantly above 0 (Exemplar: $t(1269) = 54.6$, $P < 0.001$; Prototype: $t(1269) = 38.4$, $P < 0.001$; Progenitor: $t(1269) = 37.2$, $P < 0.001$; Local: $t(1269) = 37.3$, $P < 0.001$; and Nearest neighbor: $t(1269) = 53.4$, $P < 0.001$). This result shows that all the chaining models predict a significant amount of variance in the order of WSA.

Moving beyond the baseline comparison, we compared the performance of the various chaining models against one another. Based on Fig. 5, we expected that the Exemplar model and the Nearest neighbor model would perform significantly better than the other models, which was indeed the case (Exemplar: vs. Prototype $t(1269) = 11.0$, $P < 0.001$; vs. Progenitor $t(1269) = 8.1$, $P < 0.001$; vs. Local $t(1269) = 14.0$, $P < 0.001$. Nearest neighbor: vs. Prototype $t(1269) = 8.0$, $P < 0.001$; vs. Progenitor $t(1269) = 6.5$, $P < 0.001$; vs. Local $t(1269) = 12.6$, $P < 0.001$). Although performance was most similar between the Exemplar and Nearest neighbor models, our statistical analysis also indicated that the exemplar model provided the better explanation of the data ($t(1269) = 6.1$, $P < 0.001$). This outcome was not entirely surprising, however, as it has previously been suggested that a neighborhood model converges to the exemplar model when the size of the neighborhood increases (43). In *SI Appendix*, we show that this is indeed the case, and furthermore that the exemplar and neighborhood chaining models correlate most strongly in their predictions. Taken together, these findings support for the hypothesis that similar semantic chaining mechanisms are at play both in WSA in child language development and word sense emergence in historical language evolution.

Discussion

Our work has 1) presented a computational framework of WSA grounded in the use theory of meaning and 2) illuminated how use patterns inform the development of word senses *within* each word in early life by discovering regularities and mechanisms that unify both the ontogeny and phylogeny of word sense emergence. Our computational framework enables the automatic clustering of word senses from raw transcriptions of child and child-directed

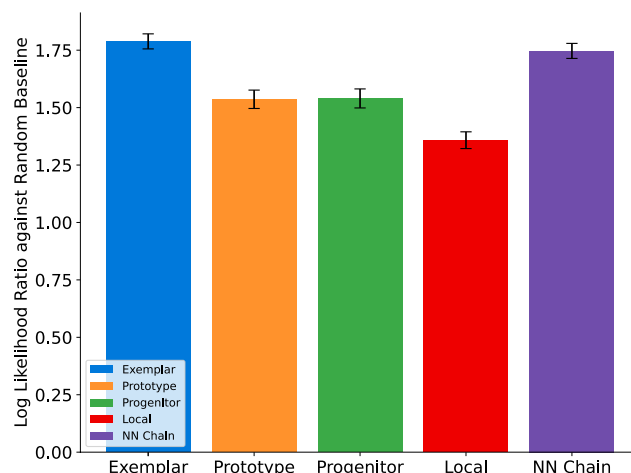


Fig. 5. Hypothesis 3: The log likelihood ratio of five semantic chaining algorithms against the random baseline. Error bars depict SEs.

speech. We illustrated the use of our framework by analyzing 1,270 words in the CHILDES corpora, showing the scalability of our approach. We have also validated the clustered senses against dictionary classifications, human ratings, and a large language model to test whether the sense clusters reflect psychologically real semantic similarities and distinctions, finding that, in general, this was indeed the case.

With this validated computational framework, we investigated why different meanings of a word come to be early in life, by testing principles of how word senses emerge. In particular, we formulated three hypotheses regarding the regular emergence of word senses that have roots in both language acquisition and language evolution—the concrete-to-abstract emergence of word senses, the incremental growth of word senses, and the mechanisms of semantic chaining. First, we found support for a concrete-to-abstract direction of sense extension as children age. Second, we found evidence that word senses acquired by children expand incrementally such that the more similar two senses are, the closer in age they are produced. Third, we found that emerging senses in children are semantically chained to previous senses. All of these findings are consistent with existing findings supporting the regularity of sense extension in historical word sense emergence (24, 25, 34, 38), and potentially reflect cognitive efficiency and least effort (56) as one important driving force in semantic change in both historical and developmental settings. As a result, our work, in conjunction with this prior work, demonstrates the unified principles and mechanisms of phylogeny and ontogeny of word meaning emergence (57).

Language Use and the Multifaceted Nature of WSA. Previous work on WSA has also been limited by the number of words and part of speech categories being examined, due to the methodological constraints of manual sense annotation (29, 30). Our work is much more comprehensive in terms of the number of words, number of utterances, and parts of speech that were examined. We found support for three important principles of WSA: concrete-to-abstract directionality, incremental sense growth, and semantic chaining in sense acquisition. Furthermore, our work has also shown nuances among the general regularities and mechanisms. For example, different parts of speech showed differential sensitivity to the pressure to acquire concrete senses before abstract senses. This directionality was present for nouns,

verbs, and adjectives—parts of speech for which individual words typically carry substantial meaning in and of themselves. However, we did not find any significant directionality of this sort for prepositions, adverbs, and pronouns (reported in [SI Appendix, section 3](#)). These findings suggest that the acquisition of word senses for different parts of speech may be shaped to different degrees by different pressures.

Our work has bolstered support for the notion that WSA is driven by a multifaceted set of pressures, including but not limited to concrete-to-abstract directionality, incremental growth, and semantic chaining. For example, previous work based on more limited data has suggested that sense frequency is also an important driving force of WSA (29, 30), in that the senses produced by children mirrored the distribution of the senses produced by their caretakers. These studies found a positive correlation between sense frequency in child speech and that in child-directed speech. In our analysis, we also found a positive correlation between children's frequency and caretakers' frequency for a given sense ($r(7644) = 0.77$, $P < 0.001$). Beyond the three hypotheses of WSA that we have examined, we also acknowledge that there are other principles of WSA that have been proposed in the literature, such as a temporal-to-spatial precedence in the emergence of senses (29, 58, 59). Also, interestingly, we did not find support for the concrete-to-abstract direction in child-directed speech. Therefore, caregivers may not choose word senses that are more concrete when talking to younger children. This finding suggests that concrete-to-abstract direction in WSA is not entirely driven by the use pattern in the linguistic environment, but potentially due to the influence of other modalities such as the visual environment or cognitive constraints during development (60, 61), such as the late development of executive function (62) that impacts the order in which senses are acquired. In sum, we view our work not as having identified the only factors that influence WSA, but as providing a vastly stronger foundation for evaluating a broad set of principles of WSA at scale and in natural, child-focused language use.

Connecting Ontogeny and Phylogeny in Word Sense Emergence.

A broad body of evidence has suggested a link between phylogeny (i.e., language evolution over historical time) and ontogeny (i.e., language development over an individual's lifespan), with shared regularities in language change across both timescales (57, 63, 64). However, an open question is whether the emergence of new senses during child language acquisition specifically also shares underlying principles and mechanisms to analogous findings in historical language evolution research. The appeal and possible value of such a unifying approach is illustrated in [Fig. 1](#), which shows how the *thin piece* sense of “stick” was attested in Old English (before 1150), whereas the “adhering” sense was attested centuries later [1350 according to the Oxford English Dictionary (65)], paralleling the order in which these senses emerge in developmental time. To provide an initial substantiation of such a unifying approach, the three hypotheses examined in the current work were all inspired by analogous work in historical language evolution research that has established how concrete-to-abstract directionality (24, 35), incremental growth (36–39), and semantic chaining (23, 25, 42) constrain word sense emergence.

The fact that we found evidence supporting each of our three phylogenically inspired hypotheses in an ontogenetic context thus provides an initial demonstration that word sense emergence on both of these time scales may be shaped by similar principles and

mechanisms. Furthermore, this initial work adds nuance to the notion that “the same” principles and mechanisms are at play in both cases. For instance, it has previously been argued that WSA does not necessarily follow the concrete-to-abstract direction in developmental WSA (29, 30). Our results provide an updated and more nuanced take on this discrepancy, showing that it may be limited to certain parts of speech, namely function words, for which the concrete-to-abstract pattern exists in historical evolution as grammaticalization (e.g., “back” as a body part later grammaticalized to mean a spatial relationship, cf. 66). In contrast, a closer parallel does exist for the majority of words, including content words such as nouns, verbs, and adjectives. Our analysis of WSA with respect to different parts of speech thus contributes to the broader vision of a more systematic and detailed comparison between ontogeny and phylogeny that seeks to understand exactly where parallels exist across these domains. These comparisons may ultimately reveal how and why particular principles and mechanisms may underpin both parallels and discrepancies across phylogenetic and ontogenetic time-scales.

One Framework, Many Possibilities. We illustrated the value of our automatic and scalable framework using speech transcriptions from a large set of children from the CHILDES corpus, but its potential uses are much broader. As denser longitudinal datasets emerge (67–69), it should be possible to study individual patterns of WSA in ecologically valid settings. It can also be used to study how older children are exposed to new senses through reading [e.g., by analyzing the CYP-LEX dataset of books read by 7 to 13+ year old children (70)], or from movie and television subtitles (71). Going beyond text and speech, our framework can be applied to joint multimodal embeddings [e.g., between text and vision (67–69)] to tease apart how visual input impacts WSA (see also ref. 72). Such an approach could also consider the role of joint attention between a child and caregiver modulates sense acquisition (7, 73).

Going beyond WSA, our framework can also provide an important resource for identifying senses that does not rely upon dictionaries or extensive participant-based norming, each of which has major limitations in terms of their validity and scalability (for a review, see ref. 74). It also highlights the potential of our approach to produce age-specific measures of homonymy and polysemy, and for testing at scale claims that young children have difficulty learning the unrelated meanings of homonyms (75–81) compared to polysemes (27, 82–84). Indeed, the present work has already shown that a large number of words that are homonymous to adults are only polysemous to children. More broadly, although our initial work has focused on English, it is readily applicable to other languages, thereby overcoming Anglocentrism in language research (85), as well as to target populations (e.g., based on age, geography, etc.) sampled at different points in time, thereby enabling the identification of new senses as they emerge in diverse contexts.

Conclusion

How children acquire multiple senses of words in the lexicon is a fundamental question that ties into current advances, as well as current challenges, faced by researchers in philosophy, linguistics, psychology, and computer science. Utilizing a multidisciplinary approach, we developed a computational framework based on the use theory of semantics to automatically induce word sense clusters from child and caretaker speech at the scale of the lexicon and then identified regularities and mechanisms in word sense acquisition during child language development. We found that

1) word sense extension tends to occur from concrete senses to more abstract senses, 2) senses tend to extend incrementally as children age, and 3) new senses are chained to semantically related senses that children have already learned. These regularities and mechanisms point to a shared cognitive foundation for word sense emergence in ontogeny and phylogeny.

Materials and Methods

Dataset. To demonstrate the scalability of our computational approach and provide a much more extensive representation of the lexicon than in prior work on child word sense acquisition, we first collected the most frequent 2,000 words in the North American English section of the CHILDES corpus (48). We then filtered out 485 words that were not included in the Wordsmyth dictionary. These words were mostly abbreviated words such as “don’t” and proper names such as “Carl.” We also added to our sample 12 additional words from the MacArthur-Bates Communicative Development Inventories (86) that were not already in the set (e.g., “aunt,” “babysitter”), so as to increase the relevance of our dataset to the research on child development. As a final step, we removed 143 words that were interjections, pure letters (e.g., “b,” “c”), babbling, and proper nouns. In total, this procedure left us with 1,384 words for utterance sampling.

Next, we sampled all the utterances in the North American English section of the CHILDES corpus containing the aforementioned words. An utterance is defined as an uninterrupted sequence of sentences by the same speaker as shown in Table 1, so we included utterances both from children and directed to children. There are in total 4,594,719 utterances for these 1,384 words and on average 3,320 utterances for each word. To reduce the influence of unreliable clustering for less frequent words, we removed 114 words with less than 200 utterances. This left us with 1,270 target words for our clustering and main analyses.

Embedding-Based Representations of Context. We generated embedded representations for all the utterances for the 1,270 target words using the BERT language model (15). We chose BERT because such contextualized embedding models have been demonstrated to capture word senses, as evidenced by their correlation with human judgments of sense similarities (87) and their high accuracy in word sense disambiguation (21). This model is also fully open-sourced and is small enough to be run and, if desired, retrained on relatively modest computer hardware, improving the accessibility of our work to a broader set of researchers. We input each utterance into the BERT model separately, while masking the target word. Then we averaged the last 6 hidden states of the masked target word as a context vector, as it reflects the semantic contribution to the masked target word from the context, which was expected to approximate the sense of the target word (88, 89).

Methodology for Sense Clustering. For each word, we collected all the context vectors for utterances in both child speech and child-directed speech (this is the only place we mixed the two kinds of speech in our analyses). We then distilled these vectors down to their top 2 principal components using PCA. Dimensionality reduction is necessary because prior work has shown that word senses are captured in a subspace of BERT embeddings, which are high-dimensional and prone to the curse of dimensionality (i.e., all points are far apart in high-dimensional space) (21, 43).

We input these distilled context vectors into a Dirichlet Process Gaussian Mixture Model (90) to identify sense clusters. The weight concentration prior γ was set to the default value: 1/100, and the maximal number of clusters to be identified was set to 100. The specific number of sense clusters identified for each word could be fewer than this maximum and was determined automatically based on the data for a given word.

To ensure that our findings were robust to the specific implementational choices described above, we also conducted two robustness checks. These checks were performed on a smaller set of 97 words which are the most frequently studied homonyms and polysemes in previous literature (74) due to their computational demands. In the first test, we varied the number of PCA components that we extracted from the raw BERT context vectors between 2 to 10, while holding all other aspects of the procedure constant. In the other

test, we used the KMeans algorithm (90) to cluster the context vectors instead of the Dirichlet Process Gaussian Mixture Model, holding all other procedures the same and fixing the number of PCA components to 2. Both checks yielded qualitatively similar results to those obtained with the main clustering method reported here. Details are provided in *SI Appendix*.

Human Judgments of Sense Cluster Similarity. To determine if the clusters induced from our computational frame and their underlying semantic space have some validity vis-à-vis the human mental lexicon, we ran a subjective rating experiment. This experiment was approved by University of Toronto Research Ethics Board (Protocol #: 48245). In this experiment, 183 participants were asked to rate the meaning similarity of the same target word embedded in two utterances for a set of 93 words, which were among the most studied ambiguous words in the literature (74). All participants provided informed consent prior to completing the experiment. Ratings were entered on a standard 1-7 Likert scale (1 = very different meanings, 7 = very similar meanings). The two utterances could come from three conditions: the same sense cluster, different but closely related sense clusters, and different and distantly related sense clusters. We preregistered the experiment on the Open Science Framework, and our prediction was that rated similarity would be highest among utterances in the same sense cluster, followed by different but closely related sense clusters, and lowest among utterances from different and distantly related sense clusters. See *SI Appendix* for details.

Analysis for the Three Hypotheses.

Hypothesis 1. We collected all the sense clusters and calculated the following measures to examine the hypothesis of concrete-to-abstract emergence of word senses.

Sense age. The average age associated with all the utterances of child speech for each sense cluster.

Sense concreteness. The average concreteness associated with all the utterances for each sense cluster. The concreteness for each utterance was estimated using a recently developed and validated procedure (91), which itself was based on human norms of word concreteness (92).

Log sense frequency. For each sense cluster, its frequency was calculated as $\log_{10}(\text{UtteranceCount} + 1)$, the same nonlinear transform typically applied to word-level frequency data (71). *UtteranceCount* is the number of utterances produced in child-directed speech belonging to each sense cluster. This variable represents the frequency of senses in the environment where word sense acquisition occurs. Inclusion of this variable rules out a potential confound and an alternative explanation for our findings that has been raised in prior work on a small set of words (29).

Noun proportion. The proportion of utterances in which the target word was used as a noun across all utterances. The noun labels come from the CHILDES dataset. This control variable was included to rule out a potential confound: that noun senses are acquired earlier than verb senses (54).

We used a mixed-effects regression model to predict the average age of each sense in child speech. Similarly to the model used to test Hypothesis 1, we included three fixed effects: 1) sense concreteness, 2) log sense frequency in child-directed speech, and 3) noun proportion in each sense. We also included by-item random intercepts and by-item random slopes for the predictors by fitting a maximally feasible model using *buildmer* (93). The analysis was also repeated for nouns, verbs, and adjectives. The part of speech labels were derived for each word in the following way: For each word, we collected all POS labels for this target word in each utterance and used the most frequent POS label as the POS label for this word.

Hypothesis 2. For the hypothesis concerning incremental growth of word senses, we first collected all the sense cluster pairs (i.e., the pair-wise combinations of sense clusters) for the 1,270 words. Two words, “pond” and “splash,” were then removed because after sense clustering they had only one sense that had an available mean age in child speech (the other sense appeared only in child-directed speech), such that age difference in child speech could not be calculated for them. Next, we calculated the following measures for each pair of sense clusters as the response variables and predictors in the regression analysis.

Age difference. The absolute difference between the mean age of the utterances in each of the two sense clusters in child speech. Due to skewness in this measure, we applied a logarithmic transformation to the raw age difference.

Sense distance. The semantic distance between the mean context vectors for the two sense clusters, as measured by the cosine distance of these two mean context vectors in the original BERT hidden space.

Log sense frequency difference. The absolute difference between the sense frequencies of the two sense clusters in the CHILDES corpus. As described above, sense frequency was calculated as $\log_{10}(\text{UtteranceCount} + 1)$ and represents the frequency of the sense in the child's environment.

We conducted regression analysis by fitting a mixed-effects model with all the pair-wise combinations of sense clusters. This model predicted the age difference between each pair of sense clusters of a given word in child speech from two fixed effect predictors: 1) sense distance 2) log sense frequency difference. We also included by-item random intercepts and by-item random slopes for the predictors by fitting a maximally feasible model using *buildmer* (93). Besides applying the regression model to our aggregate set of words, we also fitted 3 separate models for the nouns, verbs, and adjectives.

Hypothesis 3. The semantic chaining algorithms used in ref. 25 model the probability of a sequence of emerging word senses $s_{1:t}$ (i.e., $\{s_1, s_2 \dots s_t\}$). This probability is factored into a set of conditional probabilities modeling the sequence of emerging senses from the initial point to the end point of time observed for a given word: $P(s_{1:t}) = \prod_{i=1}^t P(s_i | s_{1:i-1})$, with $P(s_1 | s_{1:0}) \equiv P(s_1)$. In the context of word sense emergence in language development, the conditional probability of a new sense s^* being the next acquired sense given the set of already emerged senses $s_{1:t}$, $P(s^* | s_{1:t})$, predicts which sense will likely emerge next given the senses that have already emerged. The conditional probability function is based on a semantic similarity measurement $\text{sim}(s_a, s_b)$ that quantifies the semantic similarity of two senses. Similar to the clustering

approach, we first reduced the context vectors for each word to their first two principal components and used the mean of the context vectors in each cluster as the sense vector. Semantic similarity between two senses was then calculated as the exponentiated negative cosine distance between the two sense vectors. With this similarity function, the five semantic chaining models are defined as follows:

Exemplar. the probability of s^* being the next learned sense is proportional to its average semantic similarity to all learned senses: $P(s^* | s_{1:t}) \propto E_{s \in s_{1:t}} [\text{sim}(s^*, s)]$.

Prototype. the prototype of $s_{1:t}$ is the sense that is most similar to all other learned senses: $\max_{s_i \in s_{1:t}} \sum_{j \neq i} \text{sim}(s_i, s_j)$. The probability of s^* being the next learned sense is proportional to its semantic similarity to the prototype of all learned senses: $P(s^* | s_{1:t}) \propto \text{sim}(s^*, \text{Prototype}(s_{1:t}))$.

Progenitor. the probability of s^* being the next learned sense is proportional to its semantic similarity to the earliest learned sense: $P(s^* | s_{1:t}) \propto \text{sim}(s^*, s_1)$.

Local. the probability of s^* being the next learned sense is proportional to its semantic similarity to the latest learned sense: $P(s^* | s_{1:t}) \propto \text{sim}(s^*, s_t)$.

Nearest neighbor. the probability of s^* being the next learned sense is proportional to its semantic similarity to the most similar sense among the learned senses. $P(s^* | s_{1:t}) \propto \max_{s \in s_{1:t}} \text{sim}(s^*, s)$.

Data, Materials, and Software Availability. Code and data have been deposited in OSF (<https://osf.io/86yxzl/>) (94).

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1. J. Rodd, G. Gaskell, W. Marslen-Wilson, Making sense of semantic ambiguity: Semantic competition in lexical access. *J. Mem. Lang.* **46**, 245–266 (2002).
2. J. Pustejovsky, *The Generative Lexicon* (MIT Press, 1998).
3. C. M. Brugman, *The Story of Over: Polysemy, Semantics, and the Structure of the Lexicon* (Taylor & Francis, 1988).
4. M. Bréal, *Essai de sémantique (science des significations)* (Hachette, 1904).
5. F. de Saussure, *Course in General Linguistics* C. Bally, A. Sechehaye, A. Riedlinger, Eds. (Duckworth, London, 1983).
6. S. Carey, E. Bartlett, "Acquiring a single new word" in *Proceedings of the Stanford Child Language Conference* (Stanford University Department of Linguistics, 1978), vol. 15, pp. 17–29.
7. D. A. Baldwin, Infants' ability to consult the speaker for clues to word reference. *J. Child Lang.* **20**, 395–418 (1993).
8. P. Bloom, *How Children Learn the Meanings of Words* (MIT Press, 2002).
9. E. M. Markman, J. E. Hutchinson, Children's sensitivity to constraints on word meaning: Taxonomic versus thematic relations. *Cognit. Psychol.* **16**, 1–27 (1984).
10. M. Braginsky, D. Yurovsky, V. A. Marchman, M. C. Frank, Consistency and variability in children's word learning across languages. *Open Mind* **3**, 52–67 (2019).
11. A. Fourtassi, Y. Bian, M. C. Frank, The growth of children's semantic and phonological networks: Insight from 10 languages. *Cognit. Sci.* **44**, e12847 (2020).
12. L. Wittgenstein, *Philosophical Investigations* G. E. M. Anscombe, Ed. (Wiley-Blackwell, New York, NY, USA, 1953).
13. P. Horwich, *Meaning* (Oxford University Press, 1998).
14. G. Boleda, Distributional semantics and linguistic theory. *Annu. Rev. Linguist.* **6**, 213–234 (2020).
15. J. Devlin, M. W. Chang, K. Lee, K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding" in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, J. Burstein, C. Doran, T. Solorio, Eds. (Association for Computational Linguistics, Minneapolis, MN, 2019), vol. 1, pp. 4171–4186.
16. J. Rodd, Settling into semantic space: An ambiguity-focused account of word-meaning access. *Perspect. Psychol. Sci.* **15**, 411–427 (2020).
17. J. Li, B. C. Armstrong, Probing the representational structure of regular polysemy via sense analogy questions: Insights from contextual word vectors. *Cogn. Sci.* **48**, e13416 (2024).
18. J. Li, M. F. Joannise, Word senses as clusters of meaning modulations: A computational model of polysemy. *Cogn. Sci.* **45**, e12955 (2021).
19. S. Trott, B. Bergen, Word meaning is both categorical and continuous. *Psychol. Rev.* **130**, 1239–1261 (2023).
20. G. Wiedemann, S. Remus, A. Chawla, C. Biemann, Does BERT make any sense? Interpretable word sense disambiguation with contextualized embeddings. *arXiv [Preprint]* (2019). <http://arxiv.org/abs/1909.10430> (Accessed 14 October 2019).
21. E. Reif et al., "Visualizing and measuring the geometry of BERT" in *Advances in Neural Information Processing Systems* (Curran Associates, Inc., 2019), vol. 32.
22. T. Karidi, Y. Zhou, N. Schneider, O. Abend, V. Srikumar, "Putting words in BERT's mouth: Navigating contextualized vector spaces with pseudowords" in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, M.-F. Moens, X. Huang, L. Specia, S. W.-t. Yih, Eds. (Association for Computational Linguistics, Online and Punta Cana, Dominican Republic, 2021), pp. 10300–10313.
23. Y. Xu, T. Regier, B. C. Malt, Historical semantic chaining and efficient communication: The case of container names. *Cogn. Sci.* **40**, 2081–2094 (2016).
24. Y. Xu, B. Malt, M. Srinivasan, Evolution of word meanings through metaphorical mapping: Systematicity over the past millennium. *Cogn. Psych.* **96**, 41–53 (2017).
25. C. Ramiro, M. Srinivasan, B. C. Malt, Y. Xu, Algorithms in the historical emergence of word senses. *Proc. Natl. Acad. Sci. U.S.A.* **115**, 2323–2328 (2018).
26. M. Srinivasan, H. Rabagliati, The implications of polysemy for theories of word learning. *Child Dev. Perspect.* **15**, 148–153 (2021).
27. M. Srinivasan, J. Snedeker, Judging a book by its cover and its contents: The representation of polysemous and homophonous meanings in four-year-old children. *Cogn. Psych.* **62**, 245–272 (2011).
28. H. Rabagliati, J. Snedeker, The truth about chickens and bats: Ambiguity avoidance distinguishes types of polysemy. *Psychol. Sci.* **24**, 1354–1360 (2013).
29. S. Rice, Patterns of acquisition in the emerging mental lexicon: The case of to and for in English. *Brain Lang.* **68**, 268–276 (1999).
30. S. Rice, "Growth of a lexical network: Nine English prepositions in acquisition" in *Cognitive Approaches to Lexical Semantics*, H. Cuyckens, R. Dirven, J. R. Taylor, Eds. (Walter de Gruyter GmbH & Co. KG, D-10785 Berlin, 2003), vol. 23.
31. S. Meylan, J. Mankewitz, S. Floyd, H. Rabagliati, M. Srinivasan, "Quantifying lexical ambiguity in speech to and from English-learning children" in *Proceedings of the 43rd Annual Meeting of the Cognitive Science Society: Comparative Cognition: Animal Minds, CogSci 2021*, T. Fitch, C. Lamm, H. Leder, K. Teßmar-Raible, Eds. (Cognitive Science Society, 2021), pp. 861–867.
32. H. Jiang, M. C. Frank, V. Kulkarni, A. Fourtassi, Exploring patterns of stability and change in caregivers' word usage across early childhood. *Cognit. Sci.* **46**, e13177 (2022).
33. V. Kuperman, H. Stadthagen-Gonzalez, M. Brysbaert, Age-of-acquisition ratings for 30,000 English words. *Behav. Res. Methods* **44**, 978–990 (2012).
34. B. Winter, M. Srinivasan, Why is semantic change asymmetric? The role of concreteness and word frequency and metaphor and metonymy. *Metaphor Symb.* **37**, 39–54 (2022).
35. O. Fugikawa et al., A computational analysis of crosslinguistic regularity in semantic change. *Front. Commun.* **8**, e1136338 (2023).
36. E. C. Traugott, R. B. Dasher, *Regularity in Semantic Change* (Cambridge University Press, 2001).
37. M. Hilpert, Keeping and eye on the data: Metonymies and their patterns. *Trends Linguist. stud. monogr.* **171**, 123 (2006).
38. L. Frermann, M. Lapata, A Bayesian model of diachronic meaning change. *Trans. Assoc. Comput. Linguist.* **4**, 31–45 (2016).
39. R. Bamber, S. Mandt, "Dynamic word embeddings" in *Proceedings of the 34th International Conference on Machine Learning*, D. Precup, Y. W. The, Eds. (PMLR, 2017), pp. 380–389.

40. G. Lakoff, *Women, fire, and Dangerous Things: What Categories Reveal About the Mind* (University of Chicago Press, 1987).
41. B. C. Malt, S. A. Sloman, S. Gennari, M. Shi, Y. Wang, Knowing versus naming: Similarity and the linguistic categorization of artifacts. *J. Mem. Lang.* **40**, 230–262 (1999).
42. A. A. Habibi, C. Kemp, Y. Xu, Chaining and the growth of linguistic categories. *Cognition* **202**, 104323 (2020).
43. L. Yu, Y. Xu, Infinite mixture chaining: An efficiency-based framework for the dynamic construction of word meaning. *Open Mind* **9**, 1–24 (2025).
44. R. Q. Parks, Wordsmyth advanced dictionary (2024). <https://www.wordsmyth.net/>. Accessed 28 January 2024.
45. G. A. Miller, Wordnet: A lexical database for English. *Commun. ACM* **38**, 39–41 (1995).
46. R. Q. Parks, Wordsmyth word explorer children's dictionary: Elementary children's dictionary (for grades 3–5) (2025). www.kids.wordsmyth.net. Accessed 8 June 2025.
47. J. Culbertson, S. Gross, Are linguists better subjects? *Br. J. Philos. Sci.* **60**, 721–736 (2009).
48. B. MacWhinney, *The CHILDES Project: Tools for Analyzing Talk, Volume I: Transcription Format and Programs* (Psychology Press, New York, ed. 3, 2000).
49. D. M. Blei, M. I. Jordan, Variational inference for Dirichlet process mixtures. *Bayesian Anal.* **1**, 121–143 (2006).
50. J. R. Anderson, The adaptive nature of human categorization. *Psychol. Rev.* **98**, 409 (1991).
51. A. Alishahi, S. Stevenson, A computational model of early argument structure acquisition. *Cognit. Sci.* **32**, 789–834 (2008).
52. C. Izura, A. W. Ellis, Age of acquisition effects in word recognition and production in first and second languages. *Psicologica* **23**, 245–281 (2002).
53. K. Nelson, Structure and strategy in learning to talk. *Monogr. Soc. Res. Child Dev.* **38**, 1–135 (1973).
54. D. Gentner, On relational meaning: The acquisition of verb meaning. *Child Dev.* **49**, 988–998 (1978).
55. A. Xu, Y. Xu, Chaining and the formation of spatial semantic categories in childhood. *Proc. Annu. Meet. Cogn. Sci. Soc.* **43**, e4270 (2021).
56. O. Jespersen, *Language: Its Nature, Development and Origin* (H. Holt, 1922).
57. T. Brochhagen, G. Boleda, E. Gualdoni, Y. Xu, From language development to language evolution: A unified view of human lexical creativity. *Science* **381**, 431–436 (2023).
58. G. Lakoff, M. Johnson, *Metaphors We Live By* (University of Chicago Press, 1980).
59. E. C. Traugott, On the expression of spatio-temporal relations in language. *Univ. Hum. Lang.* **3**, e4051 (1978).
60. J. Piaget, *The Origins of Intelligence in Children* (International Universities Press, 1952).
61. S. Carey, *The Origin of Concepts* (Oxford University Press, 2011).
62. I. Dumontheil, Development of abstract thinking during childhood and adolescence: The role of rostralateral prefrontal cortex. *Dev. Cogn. Neurosci.* **10**, 57–76 (2014).
63. N. Chater, M. H. Christiansen, Language acquisition meets language evolution. *Cogn. Sci.* **34**, 1131–1157 (2010).
64. P. Monaghan, Age of acquisition predicts rate of lexical evolution. *Cognition* **133**, 530–534 (2014).
65. J. A. Simpson, E. Weiner, "[stick (v.1)]" in *Oxford English Dictionary*, M. Proffitt, J. Simpson, E. Weiner, Eds. (Oxford University Press, 2000).
66. B. Heine, U. Claudi, F. Hünemeyer, *Grammaticalization: A Conceptual Framework* (University of Chicago Press, Chicago, 1991).
67. W. K. Vong, W. Wang, A. E. Orhan, B. M. Lake, Grounded language acquisition through the eyes and ears of a single child. *Science* **383**, 504–511 (2024).
68. J. Sullivan, M. Mei, A. Perfors, E. Wojcik, M. C. Frank, SAYCam: A large, longitudinal audiovisual dataset recorded from the infant's perspective. *Open Mind* **5**, 20–29 (2021).
69. K. de Barbaro, C. M. Fausey, Ten lessons about infants' everyday experiences. *Curr. Dir. Psychol. Sci.* **31**, 28–33 (2022).
70. M. Korochkina, M. Marelli, M. Brysbaert, K. Rastle, The children and young people's books lexicon (CYP-LEX): A large-scale lexical database of books read by children and young people in the United Kingdom. *Q. J. Exp. Psychol.* **77**, 2418–2438 (2024).
71. M. Brysbaert, B. New, Moving beyond Kučera and Francis: A critical evaluation of current word frequency norms and the introduction of a new and improved word frequency measure for American English. *Behav. Res. Methods* **41**, 977–990 (2009).
72. L. Smith, C. Yu, Infants rapidly learn word-referent mappings via cross-situational statistics. *Cognition* **106**, 1558–1568 (2008).
73. M. Carpenter, K. Nagell, M. Tomasello, G. Butterworth, C. Moore, Social cognition, joint attention, and communicative competence from 9 to 15 months of age. *Monogr. Soc. Res. Child Dev.* **63**, 1–174 (1998).
74. J. Li, B. C. Armstrong, Issues of generalization from unreliable or unrepresentative stimuli: Broad lessons from lexical ambiguity. *Open Mind* **9**, 1185–1250 (2025).
75. M. M. M. Mazzocco, Children's interpretation of homonyms: A developmental study. *J. Lang.* **24**, 441–467 (1997).
76. M. Mazzocco, Developmental changes in indicators that literal interpretations of homonyms are associated with conflict. *J. Child Lang.* **26**, 393–417 (1999).
77. M. Mazzocco, G. Myers, L. Thompson, S. Desai, Possible explanations for children's literal interpretations of homonyms. *J. Child Lang.* **30**, 879–904 (2003).
78. M. J. Doherty, Children's difficulty in learning homonyms. *J. Child Lang.* **31**, 203–214 (2004).
79. A. Peters, E. Zaidel, The acquisition of homonymy. *Cognition* **8**, 187–207 (1980).
80. E. Kidd, J. Holler, Children's use of gesture to resolve lexical ambiguity. *Dev. Sci.* **12**, 903–913 (2009).
81. E. H. Wiig, M. F. Gilbert, S. H. Christian, Developmental sequences in perception and interpretation of ambiguous sentences. *Percept. Mot. Ski.* **46**, 959–969 (1978).
82. H. Rabagliati, G. F. Marcus, L. Pyllkanen, Shifting senses in lexical semantic development. *Cognition* **117**, 17–37 (2010).
83. M. Srinivasan, J. Snedeker, Polysemy and the taxonomic constraint: Children's representation of words that label multiple kinds. *Lang. Learn. Dev.* **10**, 97–128 (2014).
84. R. Zhu, Preschoolers' acquisition of producer-product metonymy. *Cogn. Dev.* **59**, 101075 (2021).
85. D. L. Share, On the Anglocentricities of current reading research and practice: The perils of overreliance on an "Outlier" orthography. *Psychol. Bull.* **134**, 584–615 (2008).
86. L. Fenson et al., *MacArthur-Bates Communicative Development Inventories: User's Guide and Technical Manual* (Brookes Publishing Co., Baltimore, MD, ed. 2, 2007).
87. S. Nair, M. Srinivasan, S. Meylan, "Contextualized word embeddings encode aspects of human-like word sense knowledge: Computation and language" in *Proceedings of the Cognitive Aspects of the Lexis on Workshop at the 28th International Conference on Computational Linguistics*, M. Zock, E. Chersoni, A. Lenci, E. Santus, Eds. (Cornell University, 2020).
88. C. M. Schuster, S. Hegelich, "From BERT's point of view: Revealing the prevailing contextual differences" in *Findings of the Association for Computational Linguistics: ACL 2022*, B. Magnini, Ed. (Association for Computational Linguistics, 2022), pp. 1120–1138.
89. Z. S. Harris, Distributional structure. *Word* **10**, 146–162 (1954).
90. F. Pedregosa et al., Scikit-learn: Machine learning in Python. *J. Mach. Learn. Res.* **12**, 2825–2830 (2011).
91. M. Aubin Le Quéré, J. N. Matias, When curiosity gaps backfire: Effects of headline concreteness on information selection decisions. *Sci Rep* **15**, 994 (2025).
92. M. Brysbaert, A. B. Warriner, V. Kuperman, Concreteness ratings for 40 thousand generally known English word lemmas. *Behav. Res. Methods* **46**, 904–911 (2014).
93. C. C. Voeten, buildmer: Stepwise elimination and term reordering for mixed-effects regression (R package version 2.11, CRAN, 2023).
94. J. Li, B. Armstrong, Y. Xu, Discovering regularity and mechanisms of word sense acquisition in childhood. Open Science Framework. <https://osf.io/8oyxz/overview>. Deposited 14 April 2026.